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Competition and evolution in ride-hailing market: A dynamic duopoly game model

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ABSTRACT

The platform competition in a real-world ride-hailing market is complex and dynamic. Existing research mainly relies on static competitive equilibrium, lacking an understanding of the dynamic evolution process of platform competition. This paper develops a generic duopoly evolutionary game model to capture dynamic decision adjustments and the evolution of market conditions. Bounded rationality and strategic learning are integrated into the modeling to dissect the interaction and competition between two platforms. The existence of stationary points where the system states remain unchanged over time is identified, and simplified stability conditions are derived. A pure strategy pair is found stable when constituting a strict Nash equilibrium and corresponding to positive payoffs. We establish two extended models to capture the nascent service aggregator and mixed electric/fuel fleets and compare them with the basic model. The model is calibrated employing the ride-hailing data of New York and is applicable for analyzing the market evolution under different states. The impacts of potential demand drop, aggregated services, and mixed fleets on platform decision and driver-passenger choice evolution are unveiled. It is found that the platforms possess self-recovery capability when facing a deteriorated external environment. The decision-making of the small platform exhibits a more substantial inertia and volatility since the large platform's decision can easily influence the market supply and demand. In the initial stage, the service aggregator slightly harms the large platform, but in the long run, it benefits both platforms. Besides, both platforms reduce prices while inflating commission ratios under mixed fleets. This study provides a fresh perspective for exploring the real-world dynamic ride-hailing competition and market evolution.

1. Introduction

The shared mobility market has experienced rapid growth in recent years, leading to intense competition among ride-hailing platforms worldwide. Examples include Didi and T3 in China, Lyft and Uber in the United States, and Uber and Cabify in Latin America. These platforms engage in fierce competition to gain market share, and their strategies influence each other. At the same time, to integrate transportation resources and reduce matching frictions, some mapping service providers, such as Amaps and Tencent

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Maps (Deng, 2020), and platforms like RideGuru[1] have launched aggregated ride-hailing services. These aggregators connect multiple ride-hailing platforms, allowing passengers to request services from different platforms simultaneously, thereby reducing waiting time and gaining popularity. Thus, when making decisions, platforms should consider passengers' choices for aggregators. Currently, another prominent feature of the ride-hailing market is electrification. Due to the environmental pollution caused by fuel vehicles (FVs) and government encouragement of electric vehicles (EVs), platforms such as Didi, Lyft, and Uber have gradually adopted a blended EV-FV fleet to provide ride-hailing services (Liu et al., 2022; Cai et al., 2023a). These two different driver categories of EVs and FVs greatly amplify the complexity of platform decision-making.

More importantly, platform strategies are not always naturally optimal in the real world due to bounded rationality. Instead, they dynamically update and evolve through multiple rounds of interactions and games. The platform competition, third-party aggregator, and blended fleet of EVs and FVs further intensify the uncertainty and dynamicity of platform decision-making. However, most previous studies have focused on static perspectives, investigating pricing and profits built upon the multi-platform competition equilibrium without considering the dynamic strategic adjustment process. This potential assumption of perfectly rational platforms helps gain rich managerial insights into platform decision-making in static equilibrium states. Nevertheless, it overlooks the dynamic competition process between platforms, which is crucial for profoundly understanding the highly dynamic and intricate ride-hailing market.

Therefore, this paper proposes a generic dual-platform evolutionary game dynamics framework to describe the dynamic platform competition and market state evolution. As an explicit mechanism model, it reasonably characterizes the complex evolution process in the real world. Evolutionary game dynamics has found widespread utilization in capturing the interactions among different entities in transportation issues (e.g., A. Abd et al., 2019; Encarnação et al., 2018; Wang et al., 2022) but has not been employed to model the dynamic duopoly competition between ride-hailing platforms. We approach the evolutionary game from a learning perspective rather than the traditional biological perspective (Börgers and Sarin, 1997). The evolutionary game is a process of participants' strategy learning, where two players continuously modify and update their decisions during interaction. It uses probability vectors of strategies to represent the micro-evolution of decision-making and is suitable for studying repeated games with imperfectly rational entities. Consistent with the latest situation in the ride-hailing industry, we also consider the service aggregator and hybrid EV-FV fleets in two extended models for comparison. Following the existing research on static ride-hailing platform competition (Zhang and Nie, 2021; Bernstein et al., 2021; Li et al., 2023), this paper concentrates on the dynamic game between two representative platforms. The proposed model is extensible and can be applied to explore scenarios with more platforms and other specific cases. We aim to address the following questions:

(a) How to characterize the interaction and competition between boundedly rational platforms through mechanism models?
(b) How do the service aggregator and mixed EV-FV fleets influence the dynamic duopoly ride-hailing market?
(c) How will dual-platform decisions and market conditions evolve under different exogenous environments and market structures?

To tackle these issues, we develop a basic model that considers passenger and driver choices without the aggregator and mixed fleets. A dynamic evolutionary game framework for platform strategy adjustments is constructed by incorporating the bounded rationality of platform decision-makers. Two extended models with the service aggregator and blended EV-FV fleets are developed. In the former, passengers can call services from an individual platform or concurrently request services from both platforms via the aggregator, while in the latter, two platforms employ both types of drivers. Based on the New York ride-hailing market data, we calibrate the parameters and conduct numerical experiments to investigate and compare the basic and extended models. The main contributions are summarized as follows:

(a) Methodological innovation in the dual-platform ride-hailing market research is achieved. Based on the classical theory of ride-hailing supply-demand matching, we introduce the platforms' bounded rationality and strategic learning process, providing a fresh perspective for analyzing the dynamic ride-hailing market. Moreover, the existence of stationary points is theoretically proved, and simplified stability conditions are derived.
(b) We capture two key ride-hailing market characteristics (i.e., service aggregator and mixed fleets) in two extended models. Their impacts on driver-passenger choices, platform competition, and strategy evolution are revealed by comparing them with the neat basic model.
(c) The reasonable depiction of ride-hailing platform decision-making and driver-passenger choices through an explicit mechanism model is realized. The model is calibrated using real-world data to explore the ride-hailing market evolutionary trajectory under various states, such as potential demand plunge.

The subsequent sections are organized as follows. The related literature is reviewed in **Section 2**. **Section 3** builds a neat basic model without the aggregator and mixed fleets. Then, **Section 4** depicts the platforms' strategy iteration process and analyzes the stability of the duopoly game model. Two extended models considering the above two elements are developed in **Section 5**. **Section 6** calibrates the model using real-world data and demonstrates the market evolution process. Finally, the conclusions and future works are provided in **Section 7**.

[1] Web: https://ride.guru/.

Table 1
Transportation research employing evolutionary game theory.

Reference	Subject	Summary
Encarnação et al. (2018); Kong et al. (2020)	Electric vehicle promotion	Analyze the interplay among regulators, enterprises, and purchasers and provide policy suggestions for the EV adoption dilemma.
A. Abd et al. (2019)	Traffic distribution of checkpoints	Leverage the evolutionary game theory to depict the cross-border transportation and obtain the equilibrium distribution.
Zhao et al. (2022)	Electric refrigerator van promotion	Consider the compound effects of different factors and discuss the dynamic subsidy for logistics companies and van manufacturers.
Wang et al. (2022)	Private charging facility installation	Explore the evolutionary features of property companies and car owners and propose a viable incentive scheme.

2. Literature review

This paper primarily intertwines three research directions: evolutionary game dynamics in transportation systems, competition among multiple platforms in the ride-hailing industry, and ride-hailing service aggregators.

2.1. Evolutionary game

Evolutionary game theory represents a significant theoretical breakthrough built upon classical game theory, providing a dynamic perspective on the participants' strategy updates. The applications of evolutionary game models in transportation studies are presented in Table 1. In the ride-hailing field, the market evolution modeling is still in its early stages. Considering the bounded reputation and three sub-games, Mo et al. (2023) established a stochastic evolution game framework to unveil the inherent evolutionary mechanism concerning rating, complaint, trust, and service quality. A few studies examined government regulation from an evolutionary standpoint (e.g., Wan et al., 2022; Peng et al., 2023). Notably, another dynamic framework, i.e., the day-to-day evolutionary model, has been extensively utilized in portraying the transportation system evolutionary trends (e.g., Yu et al., 2020; Xu et al., 2021; Li et al., 2022). Guo and Huang (2022) probed the driver-passenger choice evolution when two competitive platforms' decisions were predetermined. The core of the day-to-day evolutionary dynamics is the competition of travel patterns or routes for users. However, the ride-hailing market involves multiple platforms concurrently competing for drivers and passengers by adjusting their pricing strategies. This model is unsuitable for studying the higher-level evolution process of dual-platform decision-making. Therefore, we propose a generic dual-platform evolutionary game framework to portray the dynamic competition and market status evolution.

2.2. Platform competition

Amidst the prosperous expansion of the ride-hailing industry, research on platform competition has garnered increasing attention in recent years (e.g., Ni et al., 2021; Zhou et al., 2022a; Bai and Tang, 2022). Zha et al. (2016) conducted the first investigation on the duopoly ride-hailing market and unearthed that the augmented matching frictions induced by market segmentation could cause higher trip fares and lower social welfare. Expanding on this groundwork, subsequent research considered novel characteristics and structures of the competitive market. Mo et al. (2020) explored the competition equilibrium and government regulation of a blended EV-FV ride-hailing market with hybrid asset types. Cohen and Zhang (2022) investigated the collaborative services jointly offered by two competitive platforms through a generalized oligopoly market model. Li et al. (2023) incorporated both the regular price and order cancellation penalty into the competitive market modeling and dissected two opposing impacts of the penalty. Cai et al. (2023b) modeled the asymmetric competition incorporating the blended fleets and decision orders. Some studies discussed drivers' single/multi-homing behaviors. For instance, leveraging the variational inequalities, Zhang and Nie (2021) dissected the duopoly market with carpooling options and various driver-homing patterns and put forth multiple regulatory policies. Bernstein et al. (2021) analyzed the duopoly Nash equilibrium under system frictions and single/multi-homing, identifying that the complete multi-homing was detrimental to the stakeholders. From the perspective of big data, Huang et al. (2023) dissected the influences of prices and competitive intensity on market outcomes, discovering that the second-tier platforms exerted more pronounced impacts on the competition. However, most previous research concentrated on the stationary ride-hailing market, primarily based on competitive equilibrium. The dynamic interaction between platforms has seldom been considered, which does not align with the real-world highly dynamic platform competition.

2.3. Ride-hailing service aggregator

As a nascent service model in the competitive ride-hailing industry, platform aggregation has triggered the interest of academia (e.g., Pandey et al., 2019; Engelhardt et al., 2022; Zhou et al., 2022b). More recently, considering the stochastic waiting time, Bao et al. (2023) investigated the platform allocation strategy of the aggregator and pointed out that the shortest waiting time-based allocation method performed well. Based on Lagrange duality, Chen et al. (2024) introduced a heuristic approach for optimizing the ride-hailing aggregator's instant discounts and tested the algorithm employing the data from Meituan. Zhou et al. (2023) leveraged the Hotelling model to depict heterogeneous passengers and elucidated the effects of aggregated services on market metrics. They identified that the aggregation might be detrimental to highly platform-loyal passengers. These studies investigated platform aggregation from the

Table 2
Nomenclature.

Notation	Description
α, β	Unit value of travel/waiting time
c_k, C_k	Passenger utility and perceived utility of platform k, $k \in \{P, Q\}$
c_I, u_I	Passenger utility and demand of the service aggregator in the extended model I
c_u, r_d	Composite utility of choosing ride-hailing services and being a ride-hailing driver
d_k^E, d_k^F	EV/FV fleet sizes of platform k in the extended model II, $k \in \{P, Q\}$
d_k, u_k	Fleet size and demand rate of platform k, $k \in \{P, Q\}$
$\widehat{d}_k(t), \widehat{u}_k(t)$	Fleet size and demand rate at time step t
D, U	Potential driver number and passenger demand
Δt	Iteration step size
e, e_0, e_1, g	Total charging cost, monetary cost, access cost, and fuel cost
F_k, γ_k	Price and commission ratio of platform k, $k \in \{P, Q\}$
F_I	Aggregation service fee in the extended model I
$F_k(t), \gamma_k(t)$	Platform decisions at time step t
$\widehat{\phi}_k(i, j)$	Estimated profits of platform k when two platforms choose the i-th and j-th strategy in $S^{M\times 2}$ and $S^{N\times 2}$, respectively, $k \in \{P, Q\}$
$\widehat{\Phi}_k, \widehat{\boldsymbol{D}}_k, \widehat{\boldsymbol{U}}_k$	Estimated matrices of payoff, fleet size, and demand rate of platform k, $k \in \{P, Q\}$
$\widehat{\Pi}_k(t)$	Profits of platform k at time step t, $k \in \{P, Q\}$
M, N	Pure strategy pair numbers of two platforms
N^E	Charging facility scale
N_k^v	Vacant vehicle numbers of platform k, $k \in \{P, Q\}$
$\boldsymbol{p}(t), \boldsymbol{q}(t)$	Mixed strategy probability vectors of two platforms at time step t
$p_i(t), q_j(t)$	Elements of vectors $\boldsymbol{p}(t)$ and $\boldsymbol{q}(t)$
R_k	Drivers' opportunity cost of platform k, $k \in \{P, Q\}$
R_k^E, R_k^F	EV/FV drivers' opportunity costs of platform k in the extended model II, $k \in \{P, Q\}$
$S^{M\times 2}, S^{N\times 2}$	Strategy spaces of two platforms
$\boldsymbol{S}_F^{l\times 2}, \boldsymbol{S}_\gamma^{l\times 2}$	Two platforms' strategy vectors of price and commission ratio, $l \in \{M, N\}$
T	Travel time
$\theta_u^1, \theta_u^2, \theta_d^1, \theta_d^2, \theta_d^3$	Dispersion parameters
w_k, w_I	Waiting time of platform k and the aggregator, $k \in \{P, Q\}$

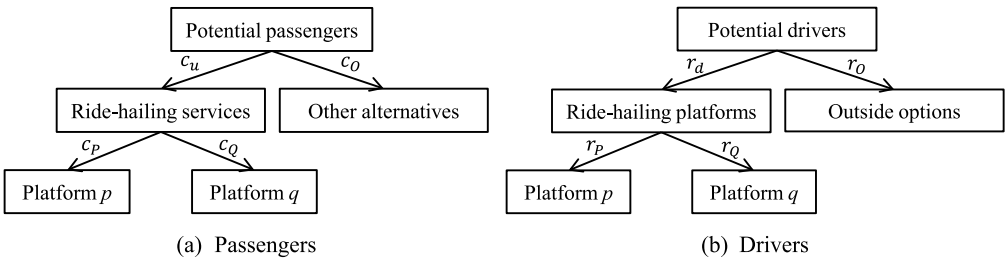

Fig. 1. Schematic diagram of passenger and driver choices.

perspectives of algorithms and equilibrium modeling. However, the effects of platform aggregation on the dynamic ride-hailing market have not been explored. Due to the dynamics and bounded rationality of decision-making, the evolutionary features of aggregated service demand remain an unresolved issue.

3. Basic model

We first consider a basic competitive market model without a service aggregator and mixed fleets. Two ride-hailing platforms, each operating a homogeneous FV fleet, compete for passengers and drivers for maximum profitability. This section delineates the lower level's demand, supply, profit, and welfare under given platform decisions. Then, in Section 4, the mutual influence of decision-making and the evolution of platform strategies are formulated. Two extended models capturing the service aggregator or mixed EV-FV fleets will be introduced in Section 5. For readability, the primary notations are summarized in Table 2.

3.1. Demand

We employ subscript $k \in \{P, Q, O\}$ to represent platform p, platform q, and other alternatives (e.g., taxis, and mass transit). C_P and C_Q denote passengers' perceived utility of services provided by the two platforms. Given the two platforms' prices F_P and F_Q, the passengers' utility of the two platforms can be written as

$$c_k = C_k - F_k - \beta w_k - \alpha T, \quad k \in \{P, Q\} \tag{1}$$

where α and β represent the unit value of travel time T and waiting time w_k, respectively.

Denote the number of unoccupied drivers on platform k as N_k^v, passengers' waiting time w_k can be expressed as

$$w_k = w(N_k^v), \quad k \in \{P, Q\} \tag{2}$$

where $w(N_k^v)$ decreases with N_k^v, that is, $\partial w(N_k^v)/\partial N_k^v < 0$.

As presented in Fig. 1(a), the passenger choices are characterized by the nested logit model. Due to internal correlation, the two platforms are grouped into one nest with a composite utility of

$$c_u = \frac{1}{\theta_u^2}\log\left(\sum\nolimits_{k\in\{P,Q\}}\exp(\theta_u^2 c_k)\right) \tag{3}$$

The utility of choosing other alternatives is denoted as c_O, which reflects the impact of other travel modes on passenger's willingness to choose ride-hailing platforms. Then, the passenger demand for the two platforms can be expressed as

$$u_k(\boldsymbol{F},\boldsymbol{\gamma}) = \frac{\exp(\theta_u^1 c_u)}{\sum_{i\in\{O,u\}}\exp(\theta_u^1 c_i)}\frac{\exp(\theta_u^2 c_k)}{\exp(\theta_u^2 c_u)}U, \quad k \in \{P, Q\} \tag{4}$$

where $\boldsymbol{F}$ and $\boldsymbol{\gamma}$ denote (F_P, F_Q) and (γ_P, γ_Q), respectively. U and θ_u^i $(i = 1, 2)$ are the potential passenger demand and dispersion parameters, respectively.

3.2. Supply

R_k represents FV drivers' opportunity cost of serving platform k $(k \in \{P, Q\})$. FV drivers' fuel cost is denoted as g. Given the two platforms' commission ratios γ_P and γ_Q, the drivers' utility on platform k can be expressed as

$$r_k = F_k(1-\gamma_k)\frac{u_k}{d_k} - g - R_k, \quad k \in \{P, Q\} \tag{5}$$

where the drivers' average earning on platform k is calculated by the total earning $F_k(1-\gamma_k)u_k$ divided by the fleet size d_k.

Considering the drivers at the occupied (i.e., $u_k T$) and pickup (i.e., $u_k w_k$) states, the number of available drivers can be determined by Little's law:

$$N_k^v = d_k - u_k(T + w_k), \quad k \in \{P, Q\} \tag{6}$$

Similar to passenger demand, driver choices, as shown in Fig. 1(b), are also depicted by the nested logit model. The composite utility of being a ride-hailing driver on platform p or q is formulated as

$$r_d = \frac{1}{\theta_d^2}\log\left(\sum\nolimits_{k\in\{P,Q\}}\exp(\theta_d^2 r_k)\right) \tag{7}$$

The number of potential drivers and the utility of choosing outside options are defined as D and r_O, respectively. Due to the widespread presence of multi-homing behavior (Yu et al., 2021), we assume that drivers are registered on the two platforms and switch between them according to the utility. Then, the fleet sizes of the two platforms can be deduced as

$$d_k(\boldsymbol{F},\boldsymbol{\gamma}) = \frac{\exp(\theta_d^1 r_d)}{\sum_{i\in\{O,d\}}\exp(\theta_d^1 r_i)}\frac{\exp(\theta_d^2 r_k)}{\exp(\theta_d^2 r_d)}D, \quad k \in \{P, Q\} \tag{8}$$

where θ_d^1 and θ_d^2 are dispersion parameters.

3.3. Profit and welfare

Under strategy pair $(\boldsymbol{F},\boldsymbol{\gamma})$, the platform profits can be expressed as

$$\Pi_k(\boldsymbol{F},\boldsymbol{\gamma}) = F_k\gamma_k u_k, \quad k \in \{P, Q\} \tag{9}$$

The social welfare, consisting of the two platforms' profits and passengers' and drivers' surpluses, is formulated as:

$$SW(\boldsymbol{F},\boldsymbol{\gamma}) = \Pi_P + \Pi_Q + PS + DS \tag{10}$$

where passenger surplus PS and driver surplus DS are deduced as follows:

$$PS = \frac{U}{\theta_u^1}\log\left(\left(\sum\nolimits_{k\in\{P,Q\}}\exp(\theta_u^2 c_k)\right)^{\theta_u^1/\theta_u^2} + \exp(\theta_u^1 c_O)\right) \tag{11}$$

$$DS = \frac{D}{\theta_d^1}\log\left(\left(\sum\nolimits_{k\in\{P,Q\}}\exp\left(\theta_d^2 r_k\right)\right)^{\theta_d^1/\theta_d^2} + \exp\left(\theta_d^1 r_O\right)\right) \tag{12}$$

4. Duopoly strategy evolution

Based on the driver-passenger choice behavior, platforms engage in price and commission ratio games on demand and supply sides. Under the traditional assumption of perfect rationality, platforms can directly reach the Nash equilibrium after a single round of game, independent of the initial market conditions. However, in the real world, due to the participant diversity and market complexity, perfect rationality and complete information are generally unattainable. Nash equilibrium is a stable state after numerous dynamic game iterations and is usually path-dependent. This is because: (a) platforms are unaware of the opponent's profitability, and there is no proactive information exchange and communication between them; (b) platforms cannot ascertain whether the other is rational, thus unilaterally adopting a Nash equilibrium strategy may incur losses. Therefore, based on the theory of evolutionary game dynamics, we quantitatively delineate the platforms' strategy iteration process at the micro level. Platforms continuously revise their strategies and strengthen the more advantageous ones during the evolutionary process. The temporal evolution of different strategies' probability density over time and the existence and stability of stationary points are analyzed.

4.1. Duopoly evolutionary game dynamics model

Due to the constraints imposed by the regulators and platforms in real-life scenarios (Zhang, 2021; Cheng, 2020), ride-hailing platforms generally cannot adopt excessively high or low pricing. Therefore, we consider bounded pure strategy spaces $S^{M\times 2}$ and $S^{N\times 2}$ of the two platforms, where $(F_P,\gamma_P)\in S^{M\times 2}$ and $\left(F_Q,\gamma_Q\right)\in S^{N\times 2}$. M and N are the numbers of strategy pairs of the two platforms, respectively. In reality, the platforms possess imperfect information regarding the market. It is posited that platform k can acquire an incomplete information set $\Omega_k=\left\{((\boldsymbol{F},\boldsymbol{\gamma}),(u_k,d_k))|(F_P,\gamma_P)\in S^{M\times 2},\left(F_Q,\gamma_Q\right)\in S^{N\times 2}\right\}$ through historical data and market surveys. Based on Eqs. (1–8), the platforms can employ methods such as maximum likelihood estimation and Bayesian inference to fit the above data and estimate parameter $\widehat{\boldsymbol{\theta}}_k$, thus completing their supply, demand, and profit under different strategy pairs:

$$\widehat{u}_k = u_k(\boldsymbol{F},\boldsymbol{\gamma}|\widehat{\boldsymbol{\theta}}_k) \tag{13}$$

$$\widehat{d}_k = d_k(\boldsymbol{F},\boldsymbol{\gamma}|\widehat{\boldsymbol{\theta}}_k) \tag{14}$$

$$\widehat{\Pi}_k = \Pi_k(\boldsymbol{F},\boldsymbol{\gamma}|\widehat{\boldsymbol{\theta}}_k) \tag{15}$$

When platform p chooses the i-th strategy in $S^{M\times 2}$ (i.e., $S^{M\times 2}(i)$) and platform q chooses the j-th strategy in $S^{N\times 2}$ (i.e., $S^{N\times 2}(j)$), their profits can be formulated as

$$\widehat{\phi}_k(i,j) = \widehat{\Pi}_k\left(S^{M\times 2}(i),S^{N\times 2}(j)\right),\ k\in\{P,Q\},\ i\in\{1,\cdots,M\},\ j\in\{1,\cdots,N\} \tag{16}$$

The platform strategy entails considering not only its profit situation but also the influence of the counterpart platform's expected strategy, forming a complex dynamic game process. To depict this process, we denote $\boldsymbol{p}$ and $\boldsymbol{q}$ as the mixed strategy probability vectors corresponding to the strategy spaces $S^{M\times 2}$ and $S^{N\times 2}$, respectively. Their elements are non-negative and satisfy $\sum_{i=1}^{M}p_i=1$ and $\sum_{j=1}^{N}q_j=1$. It can be observed that they encompass all possible decisions that the platforms may make. Probability vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ interact with each other and evolve according to their relative payoffs. Based on Tuyls et al. (2003) and Börgers and Sarin (1997), we construct a replicator dynamics framework to characterize the evolutionary game between two platforms:

$$\frac{\partial p_i(t)}{\partial t} = p_i(t)\left(\boldsymbol{e}_i^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t) - \boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t)\right) = p_i(t)G_i(\boldsymbol{p}(t),\boldsymbol{q}(t),\widehat{\Phi}_P) \tag{17}$$

$$\frac{\partial q_j(t)}{\partial t} = q_j(t)\left(\boldsymbol{e}_j^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p}(t) - \boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_Q\boldsymbol{q}(t)\right) = q_j(t)H_j(\boldsymbol{p}(t),\boldsymbol{q}(t),\widehat{\Phi}_Q) \tag{18}$$

where $\widehat{\Phi}_k$ is an $M\times N$ payoff matrix composed of $\widehat{\phi}_k(i,j)$ $(k\in\{P,Q\})$[2], and e_i is an i-th unit coordinate vector. The platforms' profit estimation $\widehat{\Phi}_k$ is subject to lag and hard to update in real time. They adjust their strategies according to Eqs. (17–18) respectively, without the knowledge of each other's payoff matrix. Taking Eq. (17) (i.e., platform p) as an example, term $\boldsymbol{e}_i^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t)$ represents the expected payoff of platform p when adopting strategy $S^{M\times 2}(i)$, and term $\boldsymbol{p}^{\mathrm{T}}(t)\widehat{\Phi}_P\boldsymbol{q}(t)$ corresponds to its average payoff of all strategies in $S^{M\times 2}$. The change rate $\frac{\partial p_i(t)}{\partial t}\frac{1}{p_i(t)}$ of strategy $S^{M\times 2}(i)$ on platform p depends on the difference between the expected earnings of this

[2] Due to the analytical complexity, this paper is built upon static profit matrices rather than evolutionary ones. The data collected during the evolutionary game are not fully utilized, which are potential resources to enhance the estimate of profit matrices when considering their learning mechanism.

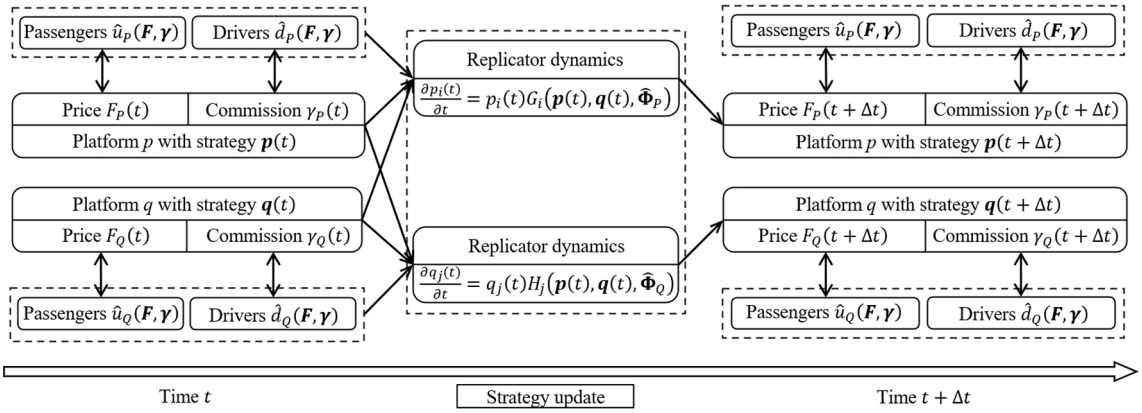

Fig. 2. Dynamic evolution and strategy updating process.

strategy and the overall average earnings. The platform will reinforce a strategy if its relative payoff is high. Notably, the expected and average payoffs of the platform's strategy are simultaneously influenced by the current strategy of the other platform.

Fig. 2 demonstrates the dynamic evolution of platforms and the strategy updating process. From a microscopic perspective, at time step t, each platform updates its probability vector for the next time step $t+\Delta t$ considering the current probabilities (i.e., $\boldsymbol{p}(t)$ or $\boldsymbol{q}(t)$), payoffs (i.e., $\boldsymbol{e}_i^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t)$ or $\boldsymbol{e}_j^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p}(t)$) of each pricing strategy (i.e., $S^{M\times 2}(i)$ or $S^{N\times 2}(j)$), and the average payoff (i.e., $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t)$ or $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_Q\boldsymbol{q}(t)$) of all strategies. In practice, the time interval Δt can be flexibly adjusted based on specific requirements, such as a day, a week, or a month.

At the macroscopic level, the platforms' decisions (i.e., $F_k(t)$ and $\gamma_k(t)$) are manifested as the weighted sum of the probability vector (i.e., $\boldsymbol{p}(t)$ or $\boldsymbol{q}(t)$) and all available strategies. That is, $F_P(t) = \boldsymbol{p}(t)^{\mathrm{T}}\boldsymbol{S}_F^{M\times 2}$, $\gamma_P(t) = \boldsymbol{p}(t)^{\mathrm{T}}\boldsymbol{S}_\gamma^{M\times 2}$, $F_Q(t) = \boldsymbol{q}(t)^{\mathrm{T}}\boldsymbol{S}_F^{N\times 2}$, and $\gamma_Q(t) = \boldsymbol{q}(t)^{\mathrm{T}}\boldsymbol{S}_\gamma^{N\times 2}$, where $\boldsymbol{S}_F^{M\times 2}$ and $\boldsymbol{S}_\gamma^{M\times 2}$ are the vectors, respectively, composed of the two columns of elements of $S^{M\times 2}$, and the definitions of $\boldsymbol{S}_F^{N\times 2}$ and $\boldsymbol{S}_\gamma^{N\times 2}$ are similar. Platform profits $\widehat{\Pi}_P(t)$ and $\widehat{\Pi}_Q(t)$ are calculated by $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}(t)$ and $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\Phi}_Q\boldsymbol{q}(t)$, respectively. Analogously, based on platform k's demand matrix $\widehat{\boldsymbol{U}}_k$ and driver supply matrix $\widehat{\boldsymbol{D}}_k$ (which correspond to the payoff matrix $\widehat{\Phi}_k$ structurally), the time-variant demand $\widehat{u}_k(t)$ and supply $\widehat{d}_k(t)$ can be expressed as $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\boldsymbol{U}}_k\boldsymbol{q}(t)$ and $\boldsymbol{p}(t)^{\mathrm{T}}\widehat{\boldsymbol{D}}_k\boldsymbol{q}(t)$, respectively. In the same manner, $\widehat{SW}(t)$ can be calculated.

4.2. Properties of stationary points

Owing to the complexity of the duopoly evolutionary game dynamics model, the theoretical analysis primarily focuses on the existence and stability of stationary points. For brevity, the following text omits the time marker "(t)". The stationary point refers to the point satisfying $\frac{\partial \boldsymbol{p}}{\partial t} = \frac{\partial \boldsymbol{q}}{\partial t} = 0$, where the system temporarily sustains a quiescent state without external disturbance but is not necessarily stable. As pointed out by Hofbauer and Sigmund (2003), the pure strategy points of symmetric games are stationary points. Specifically, for the heterogeneous game between two platforms, we introduce the following property:

Remark 1. *The system possesses at least $M \times N$ stationary points.*

The proof is available in Appendix A. Remark 1 demonstrates the existence of stationary points and provides a lower bound on their quantity, which equals the number of pure strategy pairs. However, these stationary points are not necessarily stable, as only a few may exhibit stability. The following remark presents the conditions for assessing stability based on the linear stability analysis framework.

Remark 2. *If all the eigenvalues' real parts of matrix $\begin{bmatrix} \boldsymbol{J}_{11} & \boldsymbol{J}_{12} \\ \boldsymbol{J}_{21} & \boldsymbol{J}_{22} \end{bmatrix}$ at a stationary point are negative, then the stationary point is stable. Otherwise, the point may be unstable. Here, $\boldsymbol{J}_{11}$, $\boldsymbol{J}_{12}$, $\boldsymbol{J}_{21}$, and $\boldsymbol{J}_{22}$ are defined as follows:*

$$\boldsymbol{J}_{11} = \left[\delta_{ij}\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}}\right) - p_i\boldsymbol{e}_j^{\mathrm{T}}\right]\widehat{\Phi}_P\boldsymbol{q},\ 1 \le i,j \le M \tag{19}$$

$$\boldsymbol{J}_{12} = \left[p_i\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}}\right)\right]\widehat{\Phi}_P\boldsymbol{e}_j,\ 1 \le i \le M,\ 1 \le j \le N \tag{20}$$

$$\boldsymbol{J}_{21} = \left[q_i\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}}\right)\right]\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_j,\ 1 \le i \le N,\ 1 \le j \le M \tag{21}$$

$$\boldsymbol{J}_{22} = \left[\delta_{ij}\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}}\right) - q_i\boldsymbol{e}_j^{\mathrm{T}}\right]\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p},\ 1 \le i,j \le N \tag{22}$$

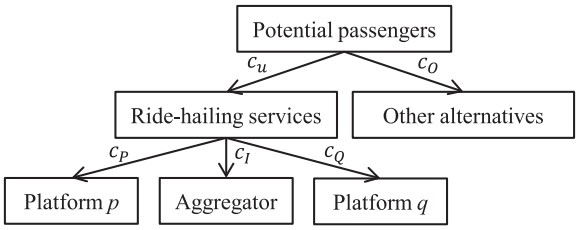

Fig. 3. Schematic diagram of passenger choices of the extended model I.

The proof is available in Appendix B. Remark 2 offers a generalized criterion to determine the stability of the stationary points. The discriminant condition is rather stringent, requiring the examination of the real parts of $M+N$ eigenvalues. It implies that only a minuscule fraction of stationary points are stable. Since the condition in Remark 2 is inconvenient to be checked, we introduce the following lemma to simplify the analysis:

Lemma 1. (Cressman, 1995). *If strategy pair $(\boldsymbol{p},\boldsymbol{q})$ is stable, it is a pure strategy pair.*

According to Lemma 1, a stable strategy must be a pure one. It also implies that a stable Nash equilibrium is pure. Therefore, the scope of Remark 2 can be narrowed down from all stationary points to all pure strategy points. We only need to discuss the stability of pure strategies:

Proposition 1. *For a pure strategy pair $(\boldsymbol{p},\boldsymbol{q})$ where $p_{\tilde{i}}=q_{\tilde{j}}=1$, if $\boldsymbol{e}_{\tilde{i}}^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{e}_{\tilde{j}}>\max\limits_{i\neq\tilde{i}}\{\boldsymbol{e}_i^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{e}_{\tilde{j}},0\}$ and $\boldsymbol{e}_{\tilde{j}}^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_{\tilde{i}}>\max\limits_{j\neq\tilde{j}}\left\{\boldsymbol{e}_j^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_{\tilde{i}},0\right\}$ hold, then the strategy is stable.*

The proof is available in Appendix C. Based on Szabó and Fath (2007), it can be summarized that for asymmetric games, the sufficient condition for stability is a strict Nash equilibrium. However, for the specific scenario of heterogeneous platform games and from the perspective of linear stability analysis, Proposition 1 reveals that the sufficient conditions for a pure strategy pair's stability are (a) constituting a strict Nash equilibrium and (b) corresponding to positive payoffs. Notably, the pure strategy pair is unstable if it yields negative payoffs for either platform. In practice, the strict Nash equilibrium ensures that the current strategy is superior to any other strategies, while the positive payoffs guarantee the rationality of the two platforms adopting the strategy. In practice, assessing the stability of an evolutionary game's terminal state can be divided into two steps. We first examine whether the corresponding strategy is a pure strategy. If yes, check whether it satisfies the above simplified conditions to determine the stability of the evolutionary endpoint eventually. Based on this, we conduct the stability assessment for the numerical examples in **Section 6**.

5. Extensions

Recently, ride-hailing service aggregators and electric vehicles have garnered popularity. Based on the basic model, we develop two extended models contemplating a service aggregator and blended EV-FV fleets. The formulation of upper-level platform evolution follows an analogous pattern to the basic model; thus, it is omitted here.

5.1. Extended model I: Service aggregator

This model primarily expands the demand side of the basic model. We use subscript I to represent the service aggregator. For passengers choosing the aggregator (i.e., request the two platforms' services simultaneously), the waiting time w_I can be expressed as

$$w_I=w\left(N_P^v+N_Q^v\right) \tag{23}$$

The probability of a passenger opting for the aggregator finally being served by platform k is proportional to the number of unoccupied drivers on platform k (Zhou et al., 2022b). We utilize λ_k to denote the ratio $N_k^v/\left(N_P^v+N_Q^v\right)$:

$$\lambda_k=N_k^v\Big/\left(N_P^v+N_Q^v\right),\qquad k\in\{P,Q\} \tag{24}$$

Therefore, the passengers' utility of choosing the aggregator can be calculated as the weighted average of the utilities of passengers served by the two platforms under platform aggregation minus the aggregator's service fee F_I:

$$c_I=\sum\nolimits_{k\in\{P,Q\}}\lambda_k(C_k-F_k-\beta w_I-\alpha T)-F_I \tag{25}$$

The passenger choices are illustrated in Fig. 3. The composite utility of choosing ride-hailing services is written as

$$c_u=\frac{1}{\theta_u^2}\log\left(\sum\nolimits_{k\in\{P,Q,I\}}\exp\left(\theta_u^2c_k\right)\right) \tag{26}$$

Then, the passenger demand for platform p, platform q, and the aggregator can be expressed as

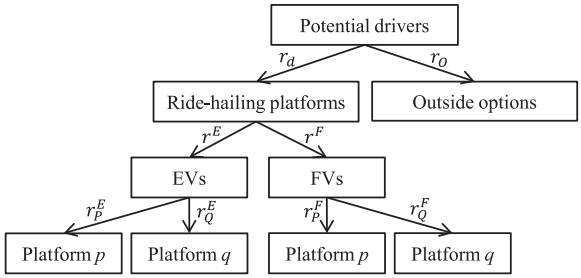

Fig. 4. Schematic diagram of driver choices of the extended model II.

$$u_k(\boldsymbol{F},\boldsymbol{\gamma}) = \frac{\exp(\theta_u^1 c_u)}{\exp(\theta_u^1 c_O) + \exp(\theta_u^1 c_u)} \frac{\exp(\theta_u^2 c_k)}{\exp(\theta_u^2 c_u)} U, \quad k \in \{P, Q, I\} \tag{27}$$

Different from Eq. (5), the drivers' utility on platform k is expressed as

$$r_k = F_k(1-\gamma_k)\frac{u_k + \lambda_k u_I}{d_k} - g - R_k, \quad k \in \{P, Q\} \tag{28}$$

where passenger demand u_k is replaced with the total demand $u_k + \lambda_k u_I$ for platform k.

The drivers can be occupied by passengers directly requesting services from platform k (i.e., u_k) and indirectly choosing platform k via the aggregator (i.e., $\lambda_k u_I$). Thus, the number of unoccupied drivers can be determined by Little's law:

$$N_k^v = d_k - u_k(T + w_k) - \lambda_k u_I(T + w_I), \quad k \in \{P, Q\} \tag{29}$$

Eqs. (1–2) and Eqs. (7–8) in the basic model are applicable to this model. Finally, the profit equals the commission portion of the price passengers pay through direct requests for services and through the aggregator:

$$\Pi_k(\boldsymbol{F},\boldsymbol{\gamma}) = F_k \gamma_k (u_k + \lambda_k u_I), \quad k \in \{P, Q\} \tag{30}$$

5.2. Extended model II: Blended EV-FV fleets

The extended model II differs from the basic model on the supply side, where drivers can choose between FVs and EVs. The demand side is defined by Eqs. (1–4). The formulation of platform profits remains consistent with the basic model (i.e., Eq. (9)). According to Ke et al. (2019), charging cost e can be written as monetary cost e_0 plus access cost $e_1\left(\left(d_P^E + d_Q^E\right)/N^E\right)$:

$$e = e_0 + e_1\left(\left(d_P^E + d_Q^E\right)\Big/N^E\right) \tag{31}$$

where e_1 rises with the two platforms' total EV fleet size $d_P^E + d_Q^E$ and diminishes with the charging facility scale N^E.

R_k^E and R_k^F represent the EV and FV drivers' heterogeneous opportunity costs of serving platform k ($k \in \{P, Q\}$). The two categories of drivers' utility on platform k can be expressed as

$$r_k^E = F_k(1-\gamma_k)\frac{u_k}{d_k^E + d_k^F} - e - R_k^E, \quad k \in \{P, Q\} \tag{32}$$

$$r_k^F = F_k(1-\gamma_k)\frac{u_k}{d_k^E + d_k^F} - g - R_k^F, \quad k \in \{P, Q\} \tag{33}$$

Similar to Eq. (6), the number of unoccupied drivers is calculated as

$$N_k^v = d_k^E + d_k^F - u_k(T + w_k), \quad k \in \{P, Q\} \tag{34}$$

Driver choices involve three levels, as illustrated in Fig. 4. The composite utility of being an EV/FV ride-hailing driver is expressed as

$$r^j = \frac{1}{\theta_d^3}\log\left(\sum\nolimits_{k\in\{P,Q\}} \exp\left(\theta_d^3 r_k^j\right)\right), \quad j \in \{E, F\} \tag{35}$$

The composite utility of being a ride-hailing driver is formulated as

$$r_d = \frac{1}{\theta_d^2}\log\left(\sum\nolimits_{j\in\{E,F\}} \exp\left(\theta_d^2 r^j\right)\right) \tag{36}$$

The EV/FV fleet sizes of the two platforms can be deduced as

$$d_k^j(\boldsymbol{F},\boldsymbol{\gamma}) = \frac{\exp(\theta_d^1 r_d)}{\sum_{i\in\{O,d\}}\exp(\theta_d^1 r_i)} \frac{\exp(\theta_d^2 r^j)}{\sum_{j'\in\{E,F\}}\exp(\theta_d^2 r^{j'})} \frac{\exp\left(\theta_d^3 r_k^j\right)}{\exp(\theta_d^3 r^j)} D, \quad (j,k) \in \{E,F\} \times \{P,Q\} \tag{37}$$

where θ_d^1, θ_d^2, and θ_d^3 are dispersion parameters.

6. Numerical experiments

Numerical experiments are conducted in this section to investigate the duopoly evolutionary game dynamics model and explore the ride-hailing market evolution process. First, we calibrate the parameters using real-world data from New York. The dynamic adjustment process of platform decisions and driver-passenger choices under the basic model is investigated. Then, we examine the market evolution trajectories under potential demand drop. Finally, the two extended models considering a service aggregator and blended EV-FV fleets are compared with the basic model.

6.1. Experimental settings and parameter calibration

We employ the fsolve function in MATLAB (R2019b) (Mathworks, 2019) and the fourth-order Runge-Kutta method (Runge, 1895) to solve the evolutionary game dynamics model of the dual-platform ride-hailing market. The solving process is presented in **Algorithm 1**. The iteration step size is set as $\Delta t = 0.004$.

Algorithm 1 Solving process of the duopoly evolutionary game model

Initialization: Strategy spaces $S^{M\times 2}$ and $S^{N\times 2}$, initial values of primary endogenous variables, and evolution starting point and step size Δt.

```
For each F and γ in strategy spaces do
    Solve the nonlinear system of Eqs. (1–8) employing the fsolve function of MATLAB.
    Calculate the platform profits.
    Record the solution as the initial value for the next iteration.
End
Establish the payoff matrixes Φ_P and Φ_Q based on the solutions.
For each iteration do
    Calculate the strategy probability vectors p(t) and q(t) using the fourth-order Runge-Kutta method.
    Obtain the time-dependent variables F_k(t), γ_k(t), u_k(t), d_k(t), and Π_k(t), where k ∈ {P, Q}.
    Update the time variable t = t + Δt.
End
```

Output: Platform decision and market state evolution trajectories.

Since this paper focuses on the evolutionary game theory of duopoly platforms, the variable values (e.g., supply and demand) in the numerical experiments are not obtained through the imputation method stated in **Section 4.1** for convenience. Instead, they are calculated using the theoretical model presented in **Section 3**, which employs real-world calibrated parameters. Specifically, this paper calibrates parameters using the New York ride-hailing market data (NYC TLC, 2023), where two primary platforms exist: Uber and Lyft. The data are processed on a weekly basis, with the last day of each week referring to that week. On April 22–28, 2019, the average price per trip for Uber was 24.12 USD, and 25.68 USD for Lyft. Since the platforms' basic pricing rules are publicly available and the prices fluctuate through coupons, incentives, and other means, we consider the pricing range for the platforms between 19.7 USD and 31.7 USD. Notice that major platforms like Uber and Didi have commission ratios of around 25% (Littleton, 2023; CICC, 2023), and governments and platforms specify the upper limits (Zhang, 2021; Cheng, 2020). Therefore, the decisions on commission ratios are constrained between 0.15 and 0.35. Indeed, setting excessively low or high prices and commission ratios would be impractical in the real world. Next, the price and commission ratio intervals are discretized into 21 specific points, respectively. Specifically, the sets of prices F_k and commission ratios γ_k are $\{19.7, 20.3, \cdots, 31.1, 31.7\}$ and $\{0.15, 0.16, \cdots, 0.34, 0.35\}$. Thus, the platforms' strategy spaces $S^{M\times 2}$ and $S^{N\times 2}$ contain 441 pure strategy pairs, i.e., $M = N = 441$.

Based on the New York ride-hailing data in 2019, we obtain an average travel time $T = 0.316$ h and an average speed of 15.58 mile/h. According to the average income of New York residents in 2019, the unit values of travel and waiting time are specified as $\alpha = \beta = 38.5$ USD/h. The fuel cost per hour g is calculated as 15.58 mile/h $\times$ 0.691 USD/L $\times$ 0.0843 L/mile $=$ 0.908 USD/h considering the average speed, New York fuel price in 2019 (EIA, 2023), and fuel consumption of a typical FV model, Toyota Camry Hybrid (CarWale, 2023). The passenger waiting time is defined by $w_k = A/\sqrt{N_k^y}$ ($k \in \{P, Q\}$), where the value of A is designated as 1 (Ke et al., 2020). The dispersion parameters are specified as $\theta_u^1 = \theta_d^1 = 0.2$ and $\theta_u^2 = \theta_d^2 = 0.1$ according to Mo et al. (2020). Passengers' and drivers' utilities of other alternatives c_O and r_O are normalized to zero.

We calibrate the remaining parameters using the New York ride-hailing data spanning four weeks from April 22 to May 19, 2019. Platform p and platform q correspond to Uber and Lyft, respectively. Due to the greater attractiveness of large platforms, the drivers' opportunity cost on platform p is lower compared to platform q, whereas the passengers' perceived utility is the opposite. Referring to New York ride-hailing drivers' hourly income (INSHUR, 2019), we set their opportunity costs of serving the two platforms (i.e., R_P and R_Q) as 22 USD/h and 28 USD/h. Passengers' relative perceived utilities per trip of the two platforms C_P and C_Q are set as 41 USD and

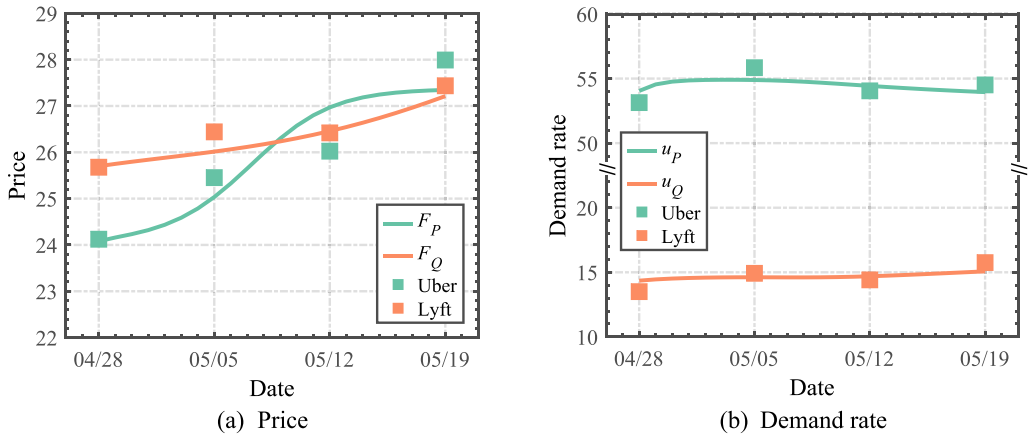

Fig. 5. Comparison between model outputs and New York ride-hailing data in 2019.

36 USD. The potential demand rate U and driver number D, as measured per unit area, are estimated as 300 /mile^2/h and 100 /mile^2 based on the realized demand and supply, respectively. The input initial strategies of platforms in the model align with the real-world data. As mentioned earlier, the average prices for Uber and Lyft on April 22–28, 2019, were 24.12 USD and 25.68 USD, respectively. The initial commission ratios for both platforms are set as 0.25. Since pure strategy pairs are the model's stationary points (Remark 1), the model stops evolving under these strategies. Therefore, we introduce a certain perturbation to the initial pure strategy pairs. Taking platform p as an example, the probability of the strategy pair (23.9, 0.25) (the closest discretized strategy to the actual pricing) is 0.9, while the probability of the other 440 strategy pairs in $S^{M\times 2}$ equals 1/4400 each. Analogously, for platform q, the probability of the strategy pair (25.7, 0.25) is 0.9. Starting from the real-world pricing data, the evolutionary trajectories and real data of the two platforms' prices and demand rates are presented in Fig. 5. It is observed that the trends of evolutionary trajectories obtained from the theoretical model are consistent with those of the real data. Meanwhile, the magnitudes of the model outputs and the real data are close, implying that the above parameter settings are reasonable.

6.2. Detailed results of the basic model

The evolution process of platform decisions, choice behaviors, and profits of the basic model is presented in Fig. 6. The dynamic adjustment of market states primarily concentrates on the first hundred time periods. It aligns with intuition, as both platforms have ample incentive to adjust their strategies to enhance profits when the market is far from equilibrium. Then, the market evolution gradually decelerates under a fixed exogenous environment. Consistent with reality, it takes several months for the market to reach a stable state preliminarily. It is observed that the strategy in the terminal state (i.e., the horizontal black dotted line) of the evolutionary game is pure. According to Proposition 1, we conclude that the market eventually attains stability since $\boldsymbol{e}_{\tilde{i}}^{\mathrm{T}}\Phi_P\boldsymbol{e}_{\tilde{j}} > \max_{i\neq\tilde{i}}\{\boldsymbol{e}_i^{\mathrm{T}}\Phi_P\boldsymbol{e}_{\tilde{j}}, 0\}$ and $\boldsymbol{e}_{\tilde{j}}^{\mathrm{T}}\Phi_Q^{\mathrm{T}}\boldsymbol{e}_{\tilde{i}} > \max_{j\neq\tilde{j}}\left\{\boldsymbol{e}_j^{\mathrm{T}}\Phi_Q^{\mathrm{T}}\boldsymbol{e}_{\tilde{i}}, 0\right\}$ hold at the endpoint.

As shown in Fig. 6(a-b), it is noted that in the evolution process, both platforms tend to inflate their prices and commission ratios. Compared to the small platform (i.e., platform q), the large platform (i.e., platform p) adopts a lower price and higher commission ratio. This corresponds to two modes. The large platform focuses on the passenger side, ensuring the demand rate through a lower price and counteracting the revenue loss caused by the low price with a higher commission ratio. The small platform, instead, concentrates on the driver side, ensuring revenue through a higher price while combining a lower commission ratio to attract more drivers. From Fig. 6(a-d), we find that pricing, supply, and demand evolutions are non-monotonic. This is because, as boundedly rational decision-makers, the platforms exhibit a certain level of inertia in their behavior, gradually attaining the optimal state through fluctuations and continuous interactions. Although the demand rate for platform p overall decreases during the evolution process, its profit is substantially enhanced (see Fig. 6(e)) due to the increased commission ratio. Generally, both platforms evolve towards directions that are more advantageous to themselves, bringing about an improvement in social welfare, as presented in Fig. 6(f).

6.3. Impacts of potential passenger demand drop

This subsection examines the influences of potential demand plunge on the market during the evolutionary process, which can be caused by exogenous factors such as epidemic control measures. The vertical black dotted line at the 30th time step represents the abrupt change in the external environment. The dotted curves (along with the superscript "b") and the solid curves represent the evolutionary trajectories under original and altered exogenous conditions, respectively.

The evolution process when potential passenger demand U drastically declines from 300 to 150 is presented in Fig. 7. After the potential demand drops, both platforms correspondingly lower their prices to attract passengers, as shown in Fig. 7(a). Differently, platform q subsequently hikes its price. The commission ratios exhibit similar trends to the prices. As illustrated in Fig. 7(b), platform p

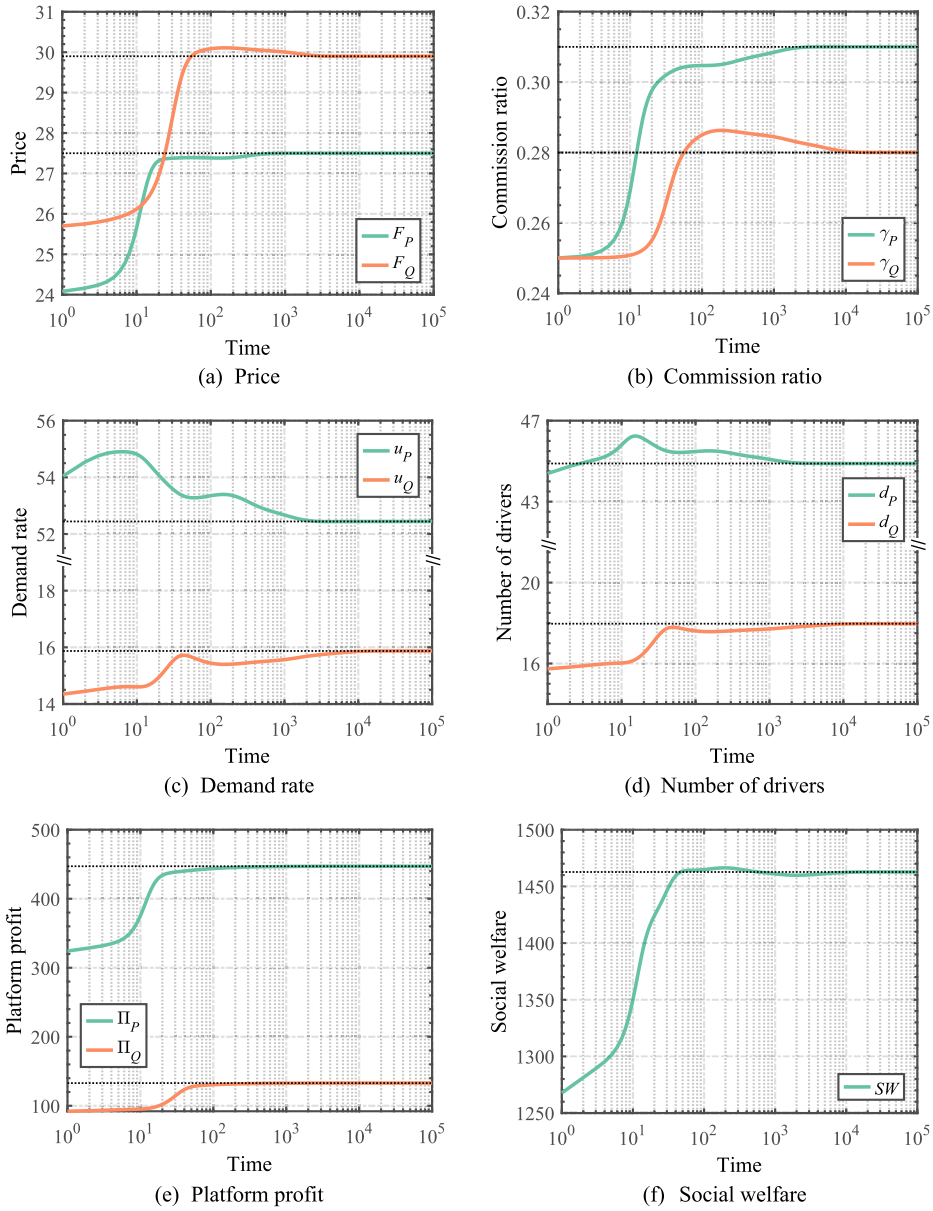

Fig. 6. Evolution process of platform decisions and market outcomes of the basic model.

constantly slashes the commission ratio, while platform q undergoes a significant fluctuation during the evolutionary process. This indicates that the decision-making of the small platform has more considerable inertia and volatility. This is because the large platform's decisions can easily impact the market regarding demand and supply, inducing considerable fluctuations in the small platform's decisions. For passenger demand, although the abrupt drop in potential demand causes a sharp decrease in platform orders, the dynamic adjustments in platform strategies gradually restore the demand and remedy the reduced profitability, as illustrated in Fig. 7(c) and (e). The model captures the platforms' strategy adjustments and self-repair capability when facing external environmental changes. Finally, with the combined recovery of driver surplus, passenger surplus, and platform profits, social welfare shows a more pronounced restoration (see Fig. 7(f)).

6.4. Comparison with extended model I

In this subsection, the market evolution under the service aggregator (i.e., extended model I) is compared with that under the basic model. The waiting time for the aggregator is specified as $w_I = A/\sqrt{N_P^v + N_Q^v}$, where the value of A is consistent with the basic model.

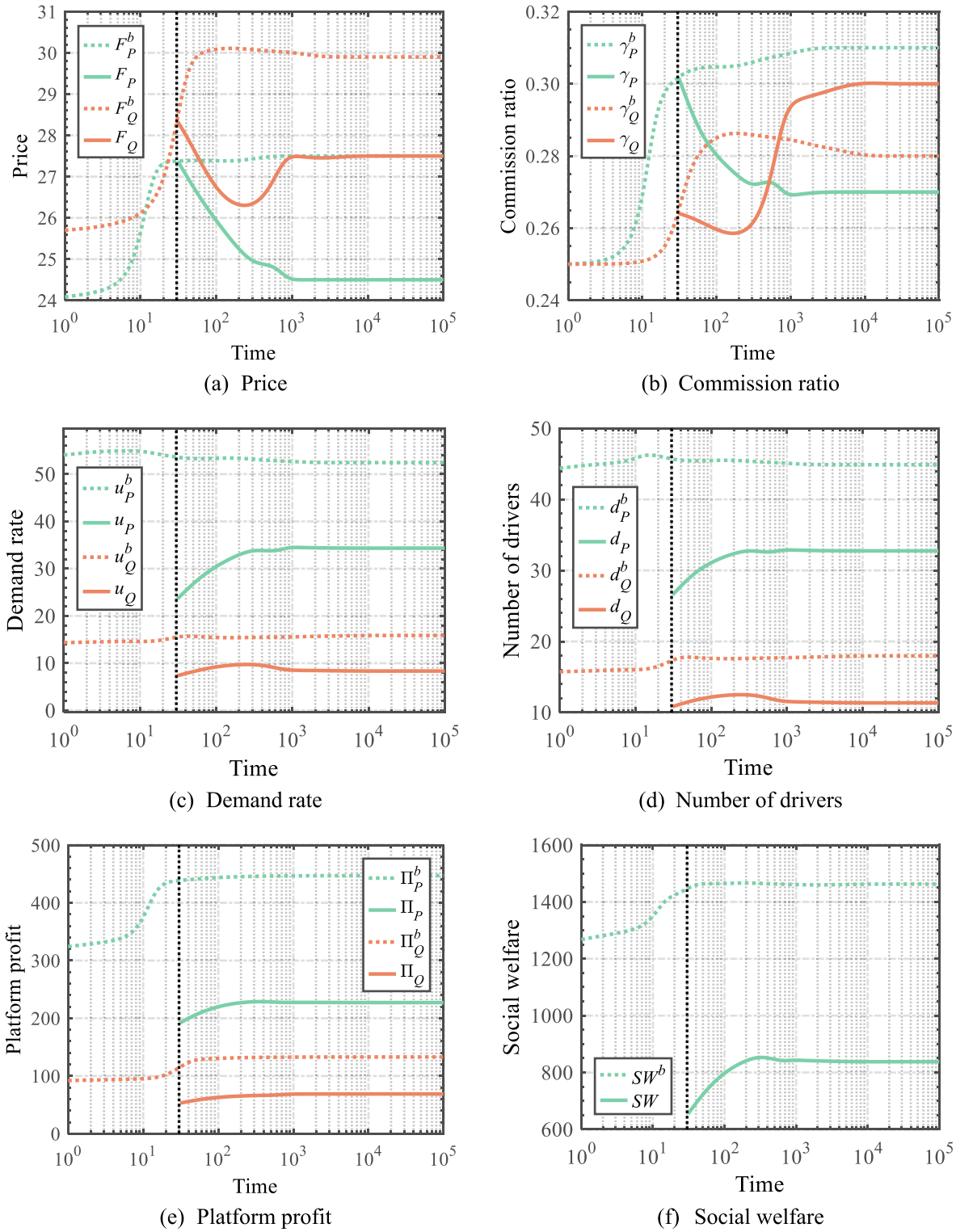

Fig. 7. Evolution process when potential passenger demand plunges.

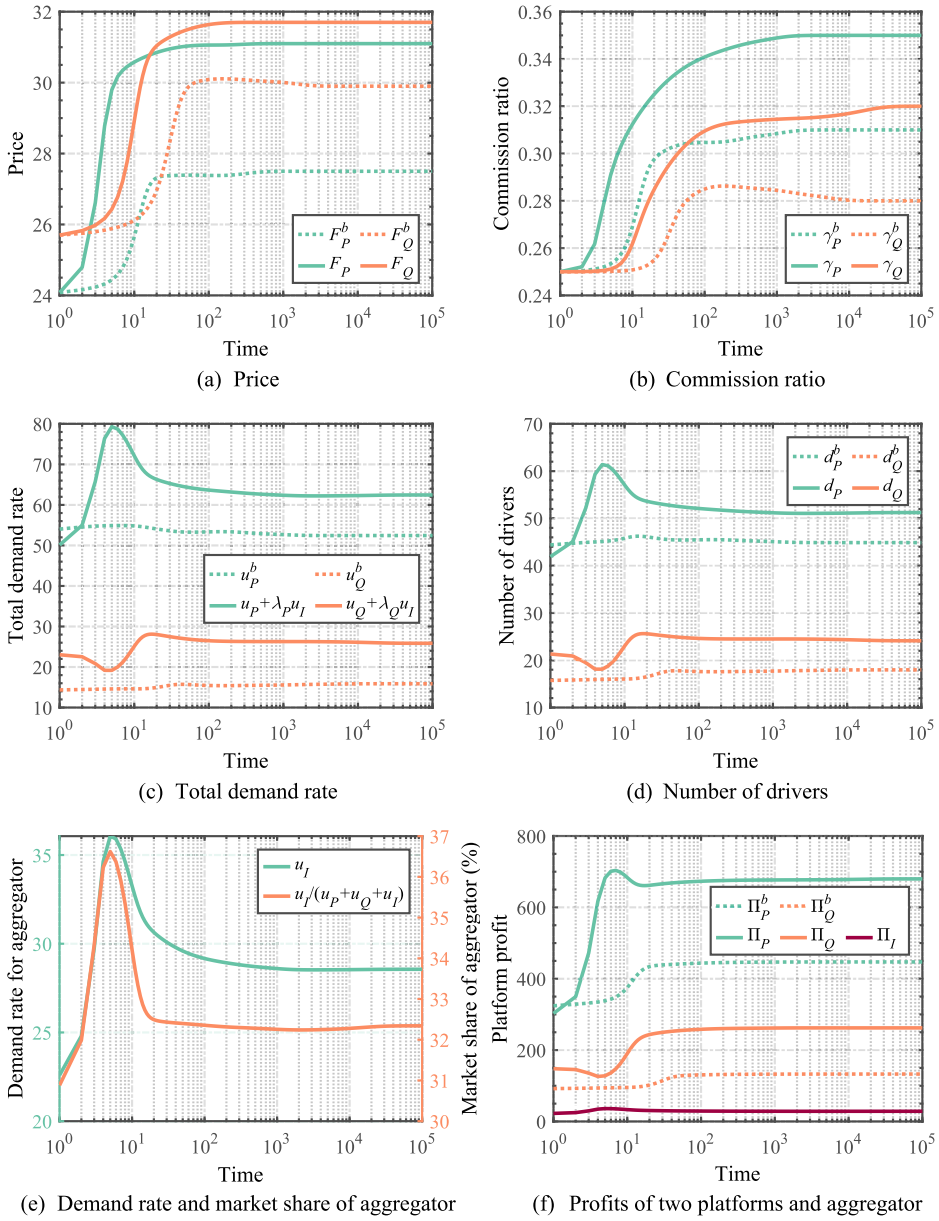

Fig. 8. Evolution process under a service aggregator.

Fig. 8 demonstrates the evolution process under an aggregator's service fee F_I of 1 USD. The superscript "b" and dotted curves represent the outcomes of the basic model.

Fig. 8(a-b) indicate that under the service aggregator, both platforms' prices and commission ratios rise faster and eventually reach higher levels than the basic model, as the platforms gain more passengers through the aggregator. Notably, Fig. 8(c), (d), and (f) show that in the initial stages, the aggregated service benefits the small platform and has a slight disadvantage for the large platform. This is because the large platform's vast driver pool results in lower waiting times for the aggregated service, allowing the small platform to "free ride" and acquire more passenger orders. As the evolution progresses, the large platform lifts the price and commission ratio to utilize the service aggregator better to boost profits. Besides, the peaks and valleys observed in Fig. 8(c-f) demonstrate the irrational factors in the evolutionary process. As presented in Fig. 8(f), the service aggregator and platform p experience short-term excess profit due to inappropriate decisions by platform q. However, in the long run, the additional returns from irrational factors will eventually converge to a rational equilibrium, giving rise to a stable state under mutual constraints.

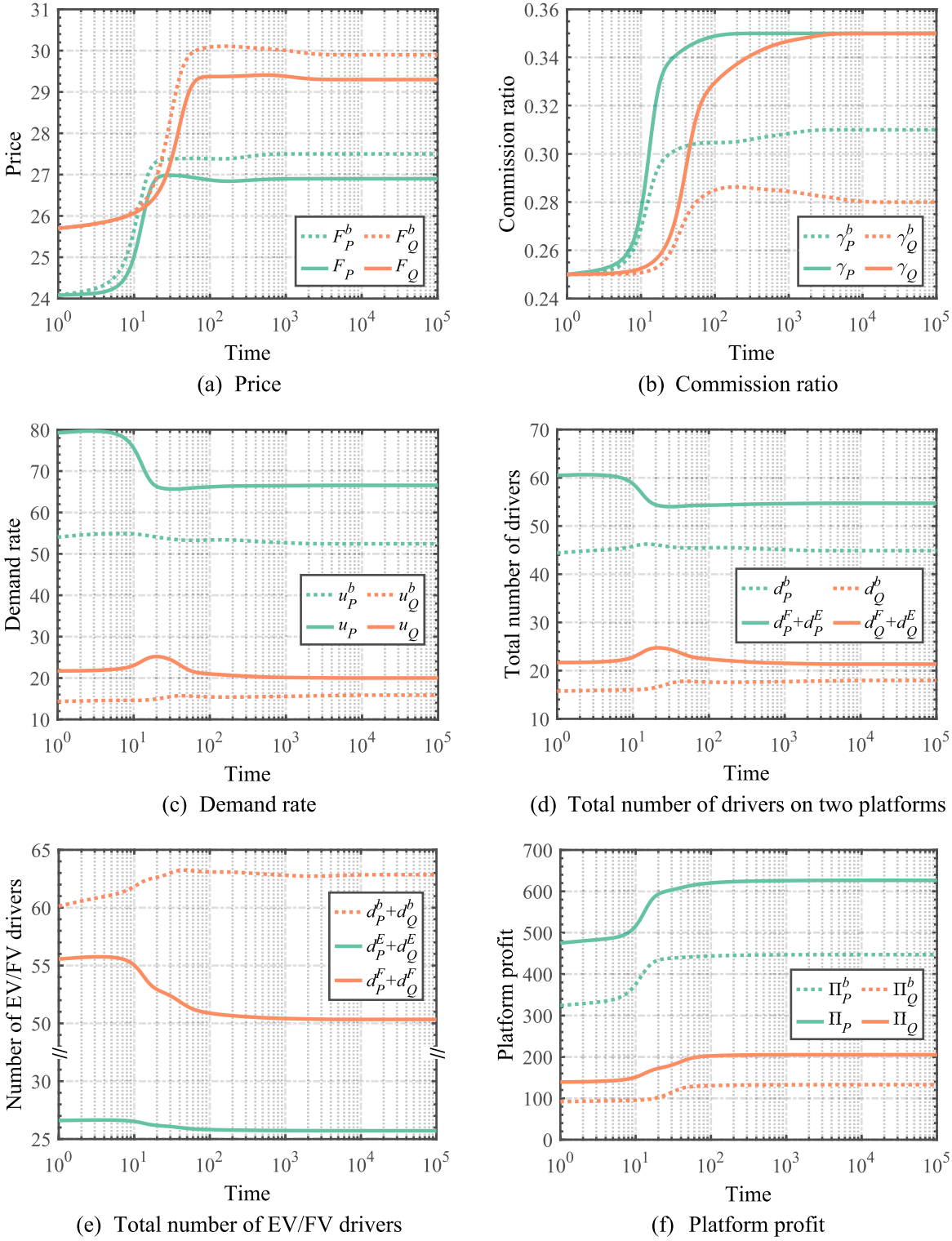

Fig. 9. Evolution process under EV/FV blended fleets.

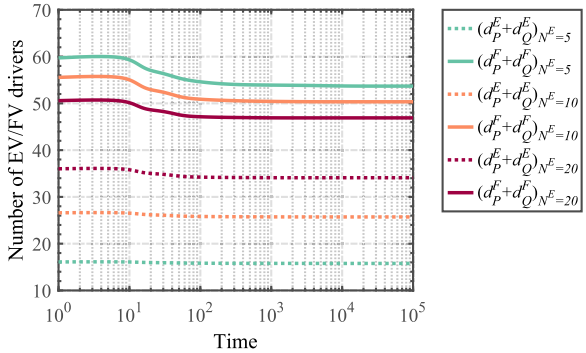

Fig. 10. Number of EV/FV drivers with respect to charging facility scale N^E.

6.5. Comparison with extended model II

We investigate the influence of blended EV-FV fleets by comparing market evolution under the basic model and extended model II, as demonstrated in Fig. 9. Analogous to the fuel cost g, the electricity monetary cost e_0 is calculated as 15.58 mile/h $\times$ 0.35 USD/kWh $\times$ 0.229 kWh/mile = 1.249 USD/h based on the New York public charging fee (NYC DOT, 2023) and electricity consumption of a common EV model, Tesla Model 3 (Electric Vehicle Database, 2023). The dispersion parameter θ_d^3 is specified as 0.1. Referring to Cai et al. (2023a), the charging access cost is formulated as $e_1 = a^E\left(\left(d_P^E + d_Q^E\right)/N^E\right)^{\psi}$, where $a^E = 0.1$, $\psi = 4$, and $N^E = 10$. Due to EVs' higher acquisition and depreciation costs, the opportunity cost of EV drivers is higher than that of FV drivers. Therefore, we set $R_P^E = 24$, $R_Q^E = 30$, $R_P^F = R_P = 22$, $R_Q^F = R_Q = 28$.

As shown in Fig. 9(a-b), under the blended fleets, the prices of both platforms decline while the commission ratios climb significantly. This is because the greater availability of drivers prompts the platforms to cut prices to attract more passengers without concerning driver wastage under excessive commission ratios. Fig. 9(e) manifests that, compared to the pure FV fleet in the basic model (the orange dotted curve), the number of FV drivers diminishes while the total number of EV and FV drivers increases under the mixed fleets. Additionally, with diverse driver sources, the platforms suppress supply during the evolutionary process, which is contrary to the basic model. Overall, the blended fleets contribute to higher platform profits, as depicted in Fig. 9(f). Fig. 10 displays the number of EV/FV drivers with respect to charging infrastructure scale N^E. Dotted curves represent the total number of EV drivers of the two platforms. The sufficiency of charging resources benefits EV drivers directly: as the number of charging facilities increases, the number of EV drivers substantially rises while the number of FV drivers declines.

7. Conclusions

This paper proposes a dual-platform evolutionary game framework to explore the dynamic competition and market state evolution. The replicator dynamics are integrated into the model to portray platform interaction and competition with bounded rationality. The existence of stationary points is identified, and sufficient conditions for a pure strategy to be stable are derived: (a) corresponding to a strict Nash equilibrium and (b) yielding positive payoffs of two platforms. Two extended models with the service aggregator and mixed EV-FV fleets are built to explore their influence on market evolution. We calibrate the model parameters resorting to the New York ride-hailing data. Numerical experiments show that the evolution of pricing, supply, and demand is non-monotonic due to the bounded rationality of decision-makers. Compared to the small platform, the large platform pays more attention to the demand side and is inclined to adopt lower prices and higher commission ratios. We further elucidate the influences of declining potential demand and introducing the service aggregator or mixed fleets on the evolution process. The results manifest that the platforms engage in strategy adjustments and profit recovery under sharply decreased potential demand, where the small platform implements higher commission ratios than before. Due to "free-riding", the service aggregator initially harms the large platform while benefiting the small one. However, as strategies evolve, both platforms benefit from it. Furthermore, during the evolution, the platforms may experience short-term unstable high profits, but a bidirectional stable profit constraint will emerge as the evolution progresses. The proposed mechanism model provides reasonable characterization for the real-world dynamic ride-hailing market.

This study offers a new perspective on platform competition research. Future research could consider expanding to the dynamic evolutionary game among platforms of arbitrary numbers. This paper assumes that the platforms implement random pricing strategies based on the probability vectors, which might differ from reality. Moreover, some more realistic learning mechanisms for platforms warrant further investigations, such as considering platforms move one gradient step, assuming the other platform does not change their pricing strategy. Bandit learning algorithm (e.g., adversarial bandits) can be implemented to model the competitive platforms and explore platform decision-making under uncertainty. Integrating the dynamic learning of drivers with the strategic evolution of platforms will provide a more realistic depiction of the ride-hailing market. Besides, our current model does not incorporate cancellation penalties and heterogeneous accuracies in the quality of provided cost/waiting data. This may result in a certain discrepancy between the model and reality. Passenger demand can be biased compared to actual demand, depending on factors such as

cancellation penalties, passenger sunk costs, and waiting times. When operating at varying supply and demand levels, different platforms may have heterogeneous accuracies in estimating travel costs and waiting times, thus affecting passenger choices.

CRediT authorship contribution statement

Zeen Cai: Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Yong Chen:** Writing – review & editing, Visualization, Validation. **Dong Mo:** Validation, Data curation. **Chaojie Liu:** Writing – review & editing. **Xiqun Michael Chen:** Conceptualization, Funding acquisition, Methodology, Supervision, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Proof of Remark 1

Consider a pure strategy pair $(\boldsymbol{p},\boldsymbol{q})$ of the two platforms. Without loss of generality, let $p_i = 1$ and $q_j = 1$. For any $k \neq i$, we have $p_k = 0$, and thus $\frac{\partial p_k}{\partial t} = p_k\left(\boldsymbol{e}_k^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q} - \boldsymbol{p}^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}\right) = 0$. Also, for any $k \neq j$, we have $q_k = 0$, and thus $\frac{\partial q_k}{\partial t} = q_k\left(\boldsymbol{e}_k^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p} - \boldsymbol{p}^{\mathrm{T}}\widehat{\Phi}_Q\boldsymbol{q}\right) = 0$. Since $\boldsymbol{p}$ and $\boldsymbol{q}$ are pure strategies, we have $\boldsymbol{p} = \boldsymbol{e}_i$ and $\boldsymbol{q} = \boldsymbol{e}_j$, and thus $\boldsymbol{e}_i^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q} = \boldsymbol{p}^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{q}$ and $\boldsymbol{e}_j^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p} = \boldsymbol{p}^{\mathrm{T}}\widehat{\Phi}_Q\boldsymbol{q}$. Therefore, for any $k \in \{1,2,\cdots,M\}$ and $k \in \{1,2,\cdots,N\}$, $\frac{\partial p_k}{\partial t} = 0$ and $\frac{\partial q_k}{\partial t} = 0$ hold, respectively, indicating that each pure strategy pair corresponds to a stationary point. There exist $M \times N$ sets of pure strategies in the strategy space $S^{M\times 2} \times S^{N\times 2}$, implying at least $M \times N$ stationary points. □

Appendix B. Proof of Remark 2

Assuming $(\boldsymbol{p}^*,\boldsymbol{q}^*)$ is a stationary point, then we have $p_i^* G_i(\boldsymbol{p}^*,\boldsymbol{q}^*,\widehat{\Phi}_P) = 0$ $(i = 1,2,\cdots,M)$ and $q_j^* H_j(\boldsymbol{p}^*,\boldsymbol{q}^*,\widehat{\Phi}_Q) = 0$ $(j = 1,2,\cdots,N)$. Following the linear stability analysis framework (Roussel, 2019) and expanding Eqs. (17–18) in the point $(\boldsymbol{p}^*,\boldsymbol{q}^*)$ by first-order Taylor series approximation yields

$$\frac{\partial p_i}{\partial t} = p_i^* G_i^* + \sum\nolimits_{k=1}^{M} \Delta p_k \frac{\partial p_i G_i}{\partial p_k}\bigg|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)} + \sum\nolimits_{k=1}^{N} \Delta q_k \frac{\partial p_i G_i}{\partial q_k}\bigg|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)} + R_G(\Delta\boldsymbol{p},\Delta\boldsymbol{q}) \tag{B1}$$

$$\frac{dq_j}{dt} = q_j^* H_j^* + \sum\nolimits_{k=1}^{M} \Delta p_k \frac{\partial q_j H_j}{\partial p_k}\bigg|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)} + \sum\nolimits_{k=1}^{N} \Delta q_k \frac{\partial q_j H_j}{\partial q_k}\bigg|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)} + R_H(\Delta\boldsymbol{p},\Delta\boldsymbol{q}) \tag{B2}$$

where Δp_k and Δq_k represent the turbulences $p_k - p_k^*$ and $q_k - q_k^*$, respectively.

Define Jacobian matrixes $\boldsymbol{J}_{11}^* = \frac{\partial(p_1G_1,\cdots,p_MG_M)}{\partial(p_1,\cdots,p_M)}\Big|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)}$, $\boldsymbol{J}_{12}^* = \frac{\partial(p_1G_1,\cdots,p_MG_M)}{\partial(q_1,\cdots,q_N)}\Big|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)}$, $\boldsymbol{J}_{21}^* = \frac{\partial(q_1H_1,\cdots,q_NH_N)}{\partial(p_1,\cdots,p_M)}\Big|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)}$, and $\boldsymbol{J}_{22}^* = \frac{\partial(q_1H_1,\cdots,q_NH_N)}{\partial(q_1,\cdots,q_N)}\Big|_{(\boldsymbol{p}^*,\boldsymbol{q}^*)}$ in the point $(\boldsymbol{p}^*,\boldsymbol{q}^*)$. Let δ_{ij} denote the Kronecker symbol, i.e., $\delta_{ij} = \begin{cases} 0, if i \neq j \\ 1, if i = j \end{cases}$. Then, the above matrixes can be concretely expressed as

$$\boldsymbol{J}_{11}^* = \left[\delta_{ij}\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{*\mathrm{T}}\right) - p_i^*\boldsymbol{e}_j^{\mathrm{T}}\right]\mathrm{diag}(\widehat{\Phi}_P\boldsymbol{q}^*),\ 1 \le i,j \le M \tag{B3}$$

$$\boldsymbol{J}_{12}^* = \left[p_i^*\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{*\mathrm{T}}\right)\right]\mathrm{diag}(\widehat{\Phi}_P\boldsymbol{e}_j),\ 1 \le i \le M,\ 1 \le j \le N \tag{B4}$$

$$\boldsymbol{J}_{21}^* = \left[q_i^*\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{q}^{*\mathrm{T}}\right)\right]\mathrm{diag}\left(\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_j\right),\ 1 \le i \le N,\ 1 \le j \le M \tag{B5}$$

$$\boldsymbol{J}_{22}^{*} = \left[\delta_{ij}\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{q}^{*\mathrm{T}}\right) - q_i^{*}\boldsymbol{e}_j^{\mathrm{T}}\right]\mathrm{diag}\left(\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p}^{*}\right),\ 1 \le i,j \le N \tag{B6}$$

Recalling $p_i^{*}G_i^{*} = q_j^{*}H_j^{*} = 0$ and neglecting the higher-order terms R_G and R_H when Δp_k and Δq_k are sufficiently tiny, we obtain

$$\frac{\partial[\Delta\boldsymbol{p}^{\mathrm{T}}\Delta\boldsymbol{q}^{\mathrm{T}}]^{\mathrm{T}}}{\partial t} = \begin{bmatrix} \boldsymbol{J}_{11}^{*} & \boldsymbol{J}_{12}^{*} \\ \boldsymbol{J}_{21}^{*} & \boldsymbol{J}_{22}^{*} \end{bmatrix}\left[\Delta\boldsymbol{p}^{\mathrm{T}}\Delta\boldsymbol{q}^{\mathrm{T}}\right]^{\mathrm{T}} \tag{B7}$$

Solving the linear ordinary differential equation systems yields

$$\left[\Delta\boldsymbol{p}^{\mathrm{T}}\Delta\boldsymbol{q}^{\mathrm{T}}\right]^{\mathrm{T}} = \sum\nolimits_{k=1}^{M+N}\eta_k e^{(\sigma_k+\omega_k i)t}\boldsymbol{v}_k = \sum\nolimits_{k=1}^{M+N}\eta_k e^{\sigma_k t}(\cos(\omega_k t) + i\sin(\omega_k t))\boldsymbol{v}_k \tag{B8}$$

where $\sigma_k + \omega_k i$ and $\boldsymbol{v}_k$ are the eigenvalue of matrix $\begin{bmatrix} \boldsymbol{J}_{11}^{*} & \boldsymbol{J}_{12}^{*} \\ \boldsymbol{J}_{21}^{*} & \boldsymbol{J}_{22}^{*} \end{bmatrix}$ and the corresponding eigenvector. η_k is an arbitrary coefficient. If all the real parts σ_k are negative, we can derive $\lim_{t\to+\infty}\left[\Delta\boldsymbol{p}^{\mathrm{T}}\Delta\boldsymbol{q}^{\mathrm{T}}\right]^{\mathrm{T}} = 0$. Thus, the impact of perturbation $(\Delta\boldsymbol{p}, \Delta\boldsymbol{q})$ on the system diminishes over time and the stationary point $(\boldsymbol{p}^{*}, \boldsymbol{q}^{*})$ is stable. □

Appendix C. Proof of Proposition 1

Denote $\begin{bmatrix} \boldsymbol{J}_{11} & \boldsymbol{J}_{12} \\ \boldsymbol{J}_{21} & \boldsymbol{J}_{22} \end{bmatrix}$ by $\boldsymbol{J}$. $\boldsymbol{J}$ is an $(M+N)$-order square matrix encompassing elements j_{kl} $(1 \le k,l \le M+N)$. Since $p_i = 0$ for any $i \ne i'$ and $q_j = 0$ for any $j \ne j'$, we have $j_{kl} = 0$ $(k \ne l, M+j', i')$. For $M+1 \le l \le M+N$ and $k = i'$ where j_{kl} is in $\boldsymbol{J}_{12}$, it can be derived that $j_{kl} = p_{i'}\left(\boldsymbol{e}_{i'}^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{e}_{l-M} = 0$. Likewise, for $1 \le l \le M$ and $k = j' + M$ where j_{kl} is in $\boldsymbol{J}_{21}$, we can obtain that $j_{kl} = q_{j'}\left(\boldsymbol{e}_{j'}^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_l = 0$. In summary, the potential non-zero elements within the matrix $\boldsymbol{J}$ are j_{kl} $(1 \le k = l \le M+N;\ k = M+j',\ M+1 \le l \le M+N;\ k = i',\ 1 \le l \le M)$. Therefore, the value of the determinant $|\boldsymbol{J} - \Lambda\boldsymbol{E}|$ is $\prod_{k=1}^{M+N}(j_{kk} - \Lambda)$, which equals

$$\prod\nolimits_{i=1}^{M}\left(\left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}} - p_i\boldsymbol{e}_i^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{q} - \Lambda\right)\prod\nolimits_{j=1}^{N}\left(\left(\boldsymbol{e}_j^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}} - q_j\boldsymbol{e}_j^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p} - \Lambda\right) \tag{C1}$$

By equating Eq. (C1) to zero, we arrive at the eigenvalues $\Lambda_i = \left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}} - p_i\boldsymbol{e}_i^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{q}$ $(1 \le i \le M)$ and $\Lambda_j = \left(\boldsymbol{e}_j^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}} - q_j\boldsymbol{e}_j^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p}$ $(1 \le j \le N)$ of the matrix $\boldsymbol{J}$. Particularly, under the premise of Proposition 1, it can be derived that $\Lambda_{i'} = \left(\boldsymbol{e}_{i'}^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}} - p_{i'}\boldsymbol{e}_{i'}^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{q} = -\boldsymbol{e}_{i'}^{\mathrm{T}}\widehat{\Phi}_P\boldsymbol{e}_{j'} < 0$ and $\Lambda_{j'} = \left(\boldsymbol{e}_{j'}^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}} - q_{j'}\boldsymbol{e}_{j'}^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p} = -\boldsymbol{e}_{j'}^{\mathrm{T}}\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_{i'} < 0$. For $i \ne i'$ and $j \ne j'$, we also have $\Lambda_i = \left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{p}^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{q} = \left(\boldsymbol{e}_i^{\mathrm{T}} - \boldsymbol{e}_{i'}^{\mathrm{T}}\right)\widehat{\Phi}_P\boldsymbol{e}_{j'} < 0$ and $\Lambda_j = \left(\boldsymbol{e}_j^{\mathrm{T}} - \boldsymbol{q}^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{p} = \left(\boldsymbol{e}_j^{\mathrm{T}} - \boldsymbol{e}_{j'}^{\mathrm{T}}\right)\widehat{\Phi}_Q^{\mathrm{T}}\boldsymbol{e}_{i'} < 0$, respectively. Recalling Remark 2, we can conclude that the strategy $(\boldsymbol{p}, \boldsymbol{q})$ is stable. □

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